

Generation Interconnection System Impact Study Report

For Queue Project Q22-2134

Calpeak Pe 7 138 kV

November 27, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2134. The Interconnection Customer proposes a 20.3 MW solar photovoltaic facility interconnecting at the Calpeak Pe 7 138 kV substation in the Virginia area. The request was received on July 14, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 632 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$198,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: HARTMAN CREEK to MOUNT VIEW 138 KV
Contingency: LOWMOOR SUBSTATION to HINTON AP 345 KV
Loading: 109.7% (Rating: 431 MVA)
DFAX: 5.5%
Pre-project loading: 109.5%
2. Facility: SAN GERMAN TC 6406 to SPAR CANON 138 KV Line
Contingency: LOCHMERE HYDROELECTRIC PLANT to PITCHGEN 138 KV
Loading: 103.5% (Rating: 178 MVA)
DFAX: 17.0%
Pre-project loading: 101.6%
3. Facility: HINTON AP to SPAR CANYON 138 KV
Contingency: PITCGEN to LOCHMERE HYDROELECTRIC PLANT 138 KV L
Loading: 101.3% (Rating: 177 MVA)
DFAX: 17.0%
Pre-project loading: 99.4%
4. Facility: SAN GERMAN TC 6406 to PITCHGEN 138 KV
Contingency: LOCHMERE HYDROELECTIC PLANT to PITCHGEN 138 KV
Loading: 100.9% (Rating: 191 MVA)
DFAX: 17.0%
Pre-project loading: 99.1%
5. Facility: FRASER to OU SPIRIT WIND FARM 138 KV
Contingency: MANATEE to HINTON AP 345 KV
Loading: 100.8% (Rating: 306 MVA)
DFAX: 17.6%
Pre-project loading: 99.6%
6. Facility: MOUNT VIEW to PEYTON 138 KV
Contingency: LOWMOOR to HINTON AP 345 KV
Loading: 100.7% (Rating: 357 MVA)

DFAX: 5.5%
Pre-project loading: 100.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HARTMAN CREEK to MOUNT VIEW 138 KV	Reconductor line	\$930,000	0% (\$4,000)
SAN GERMAN TC 6406 to SPAR CANYON 138 KV	Reconductor line	\$972,000	3% (\$29,000)
HINTON AP to SPAR CANYON 138 KV	Reconductor line	\$2,930,000	3% (\$88,000)
SAN GERMAN TC 6406 to PITCHGEN 138 KV	Reconductor line	\$1,330,000	3% (\$40,000)
FRASER to OU SPIRIT WIND FARM 138 KV	Reconductor line	\$1,972,000	1% (\$19,000)
MOUNT VIEW SUBSTATION to PEYTON 138 KV	Reconductor line	\$3,270,000	1% (\$18,000)
	Total allocated to project		\$198,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.